

QCE Chemistry Units 3&4

Test 2: Chemical equilibrium systems

- Brønsted–Lowry model
- Dissociation constants
- Acid–base indicators
- Volumetric analysis

Student's Name: _____

Teacher's Name: _____

General instructions

- Write your name and your teacher's name in the space provided above on this page.
 - A formula and data booklet is provided.
 - Unless otherwise indicated, the diagrams in this booklet are not drawn to scale.
 - Answer all questions in the spaces provided.
- Test time – 45 minutes
 - Total marks – 40 marks

SECTION 1**Instructions**

- Choose the best answer for Questions 1–10.
 - This section has 10 questions and is worth 10 marks.
 - Circle A, B, C or D.
-

QUESTION 1

Identify the equation that shows the species acting as the Brønsted–Lowry acid in **bold**.

- (A) $\text{NH}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$
- (B) $\text{H}_2\text{O}(\text{l}) + \text{CO}_3^{2-}(\text{aq}) \rightleftharpoons \text{OH}^-(\text{aq}) + \text{HCO}_3^-(\text{aq})$
- (C) $\text{HCl}(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{Cl}^-(\text{aq})$
- (D) $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{H}_2\text{O}(\text{g})$

QUESTION 2

Identify the reaction where water (H_2O) is acting as a Brønsted–Lowry base.

- (A) $\text{HF}(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{F}^-(\text{aq})$
- (B) $\text{NH}_3(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$
- (C) $\text{H}_2\text{O}(\text{l}) + \text{CO}_3^{2-}(\text{aq}) \rightleftharpoons \text{OH}^-(\text{aq}) + \text{HCO}_3^-(\text{aq})$
- (D) $\text{H}_2\text{O}(\text{l}) + \text{NH}_2^-(\text{aq}) \rightleftharpoons \text{OH}^-(\text{aq}) + \text{NH}_3(\text{aq})$

QUESTION 3

Which of the following is a weak acid?

- (A) hydrochloric acid
- (B) sulfuric acid
- (C) propanoic acid
- (D) ammonia solution

QUESTION 4

Which of the following substances is amphiprotic?

- (A) HNO_3
- (B) HCO_3^-
- (C) NaOH
- (D) NaCl

QUESTION 5

Hydrochloric acid reacts with ammonia to produce ammonium ions and chloride ions.

In this reaction, the conjugate acid is

- (A) hydrochloric acid.
- (B) ammonia.
- (C) the ammonium ion.
- (D) the chloride ion.

QUESTION 6

Which of the following solutions would have the highest hydrogen ion (H^+) concentration?

- (A) 0.4 M sulfuric acid (H_2SO_4)
- (B) 0.4 M ethanoic acid (CH_3COOH)
- (C) 0.4 M nitric acid (HNO_3)
- (D) 0.4 M sodium hydroxide (NaOH)

QUESTION 7

Which of the following would have the lowest hydrogen ion (H^+) concentration at 25°C ?

- (A) 0.1 M $\text{H}_2\text{SO}_4(\text{aq})$
- (B) 1.0 M $\text{HCl}(\text{aq})$
- (C) H_2O
- (D) 0.6 M $\text{CH}_3\text{COOH}(\text{aq})$

QUESTION 8

If a few drops of the indicator bromophenol blue were added to a solution with a pH of 2, the indicator would become

- (A) red.
- (B) yellow.
- (C) blue.
- (D) colourless.

QUESTION 9

A student performs a titration of 0.2 M ethanoic acid solution with 20 mL of approximately 0.1 M sodium hydroxide solution.

Which of the following statements is correct?

- (A) The titration would require 10 mL of ethanoic acid.
- (B) The titration would require 20 mL of ethanoic acid.
- (C) The equivalence point would be at a pH less than 7.
- (D) The equivalence point would be at a pH of 7.

QUESTION 10

Which of the following solutions would most likely make the indicator methyl orange change to red?

- (A) 0.01 M ethanoic acid
- (B) 0.01 M sodium chloride
- (C) 0.01 M sodium hydroxide
- (D) 0.01 M hydrochloric acid

SECTION 2**Instructions**

- This section has six questions and is worth 30 marks.

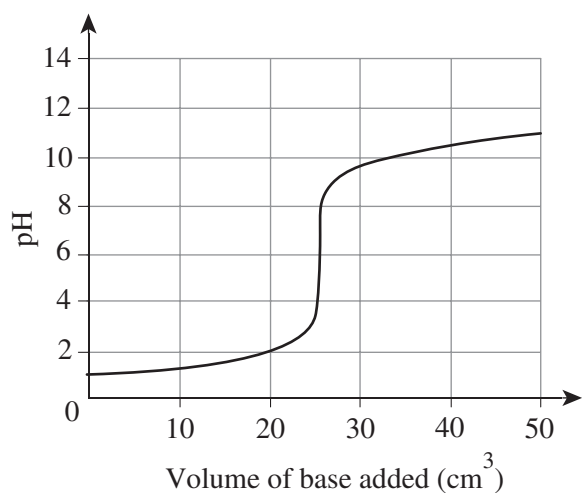
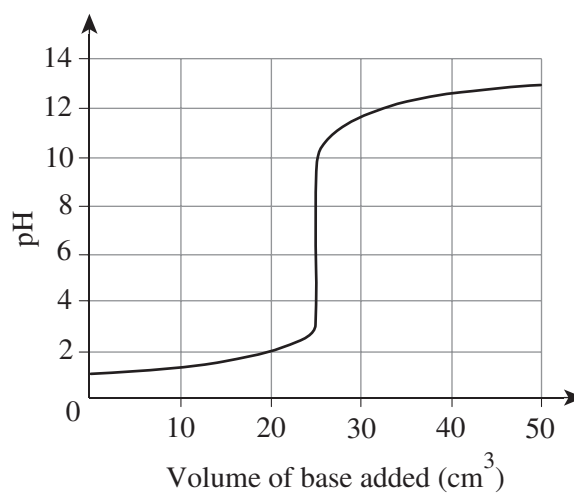
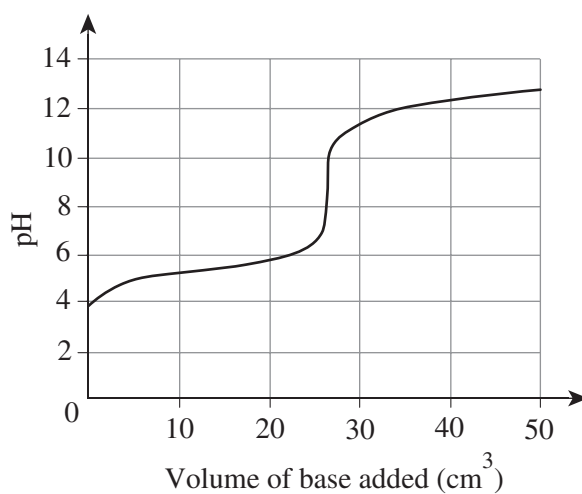
QUESTION 11 (11 marks)

A student conducted the following titrations.

- Titration 1: Hydrochloric acid with sodium hydroxide (burette)
- Titration 2: Ethanoic acid with sodium hydroxide (burette)
- Titration 3: Nitric acid with ammonia solution (burette)

Each titration used 0.1 M solutions.

Using their results, the student produced three titration curves (A, B and C), as shown.

Curve A**Curve B****Curve C**

- a) Identify which curve (A, B or C) corresponds to each titration. *[3 marks]*

Titration 1 _____

Titration 2 _____

Titration 3 _____

- b) Identify one indicator that would be suitable for titration 3. Explain your reasoning. *[2 marks]*

- c) Identify one curve (A, B or C) that indicates a buffer region and explain how the buffer works. *[3 marks]*

- d) Explain how the curve for titration 2 could be used to determine the dissociation constant (K_a) of the acid used. *[3 marks]*

QUESTION 13 (4 marks)

HCN is a weak monoprotic acid and has a dissociation constant (K_a) of 6.2×10^{-10} .

- a) Identify the conjugate base of HCN.

[1 mark]

- b) Calculate the pK_b of the conjugate base of HCN at 25°C. Show your working.

[3 marks]

$pK_b =$ _____ (to one decimal place)

QUESTION 14 (2 marks)

Acid X has a pK_a of 2.6 and acid Y has a pK_a of 4.1.

Which acid is stronger? Explain your reasoning.

QUESTION 15 (3 marks)

Ethanoic acid has a pK_a value of 4.76.

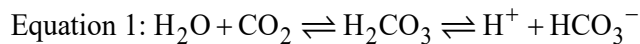
Calculate the pH of a solution containing 0.20 mol L^{-1} of ethanoic acid.

pH = _____ (to three significant figures)

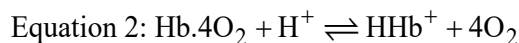
QUESTION 16 (3 marks)

In the human body, the pH of blood is maintained between 7.35 and 7.45 through a series of complex equilibria.

The first equilibrium is established between carbon dioxide, carbonic acid and hydrogen carbonate ions, according to equation 1.



The second equilibrium is established between haemoglobin and oxyhaemoglobin, according to equation 2.



The human body produces CO_2 through cellular respiration. The rate of cellular respiration and the amount of CO_2 produced increases during physical exercise.

Using the equations above, explain how the body responds to the increase in the amount of CO_2 produced in order to maintain the pH of blood.

END OF TEST